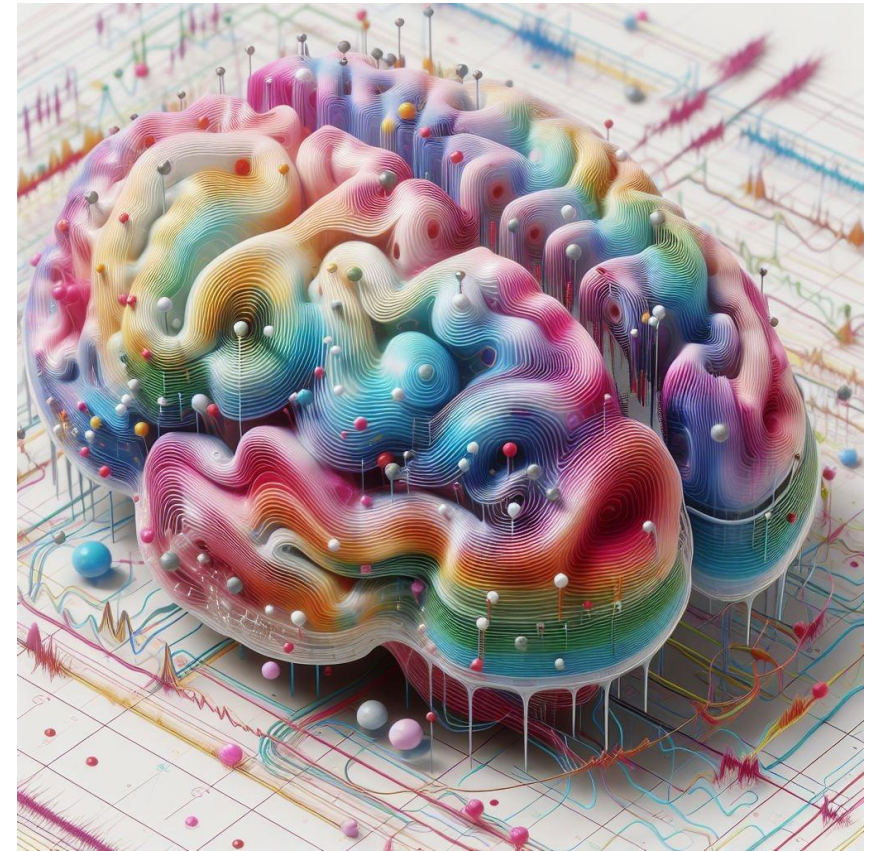




# WHOLE BRAIN MODELS OF NEURAL OSCILLATIONS

## From Local Oscillations to Large-Scale Synchronization and Entrainment

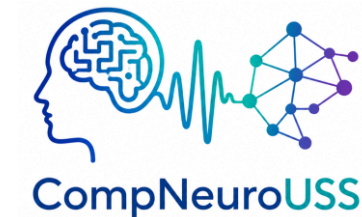
**Mónica Otero Ferreiro**

Facultad de Ingeniería,

Investigadora Asociada, Centro Basal Ciencia & Vida

Universidad San Sebastián

monica.otero@uss.cl



# WHO AM I?

## Academic and Professional Background



UNIVERSIDAD  
SAN SEBASTIAN



Fundación  
Ciencia  
& Vida



*BSc. Computer Science*, Facultad de Matemática y Computación,  
**Universidad de La Habana**, Cuba (2008)



*Programmer*, **Centro de Neurociencias de Cuba** (2008–2013)



*PhD in Electronic Engineering*, Departamento de Ingeniería Electrónica,  
**Universidad Técnica Federico Santa María**, Valparaíso, Chile (2020)



*Postdoctoral Researcher*, Escuela de Ingeniería Biomédica,  
Facultad de Ingeniería, **Universidad de Valparaíso**, Chile (2021–2022)



*Assistant Professor*, Facultad de Ingeniería, Arquitectura y Diseño,  
**Universidad San Sebastián**, Santiago de Chile, Chile (2022–presente)

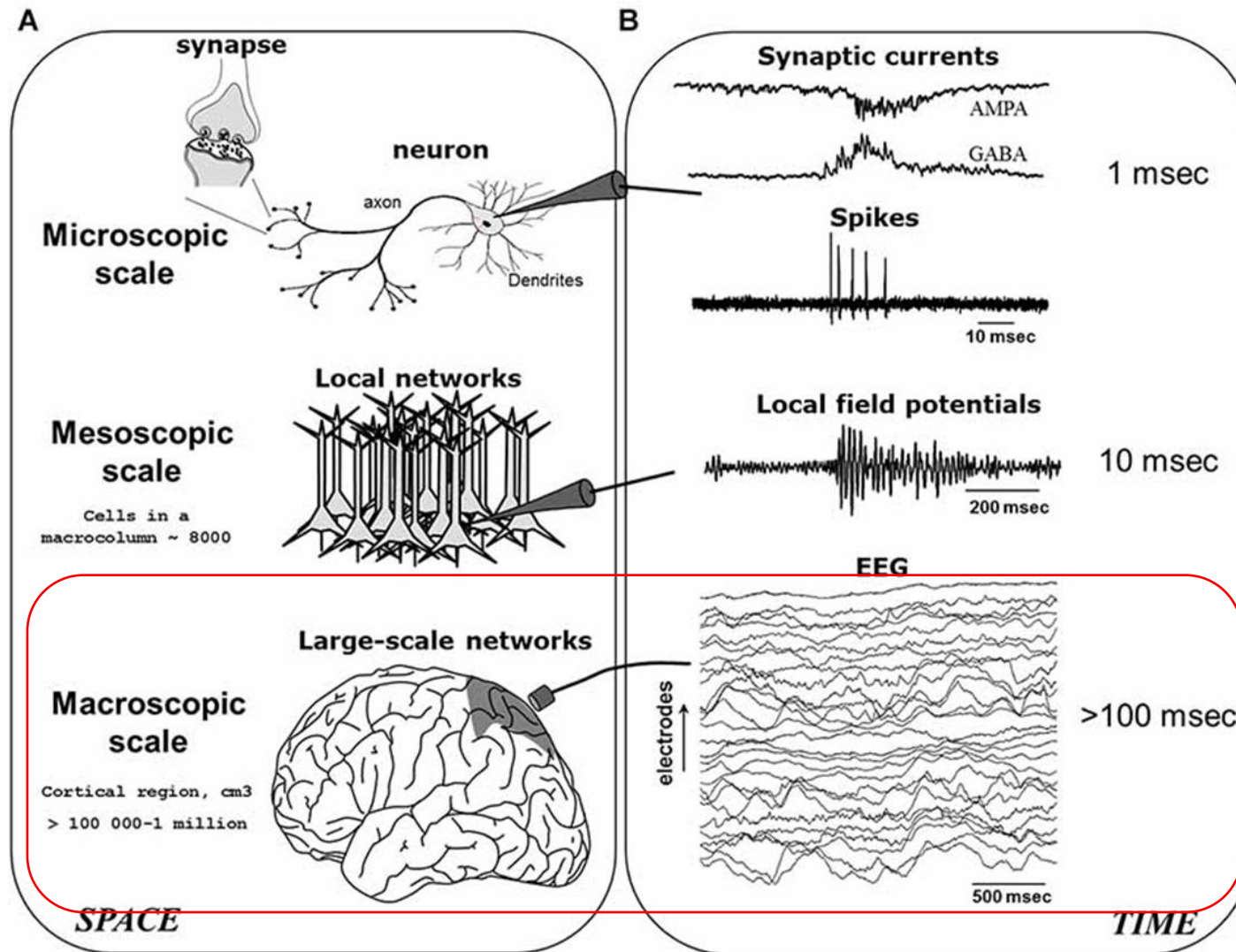


*Associate Researcher*, **Fundación Ciencia y Vida**,  
Santiago de Chile, Chile (2022–presente)

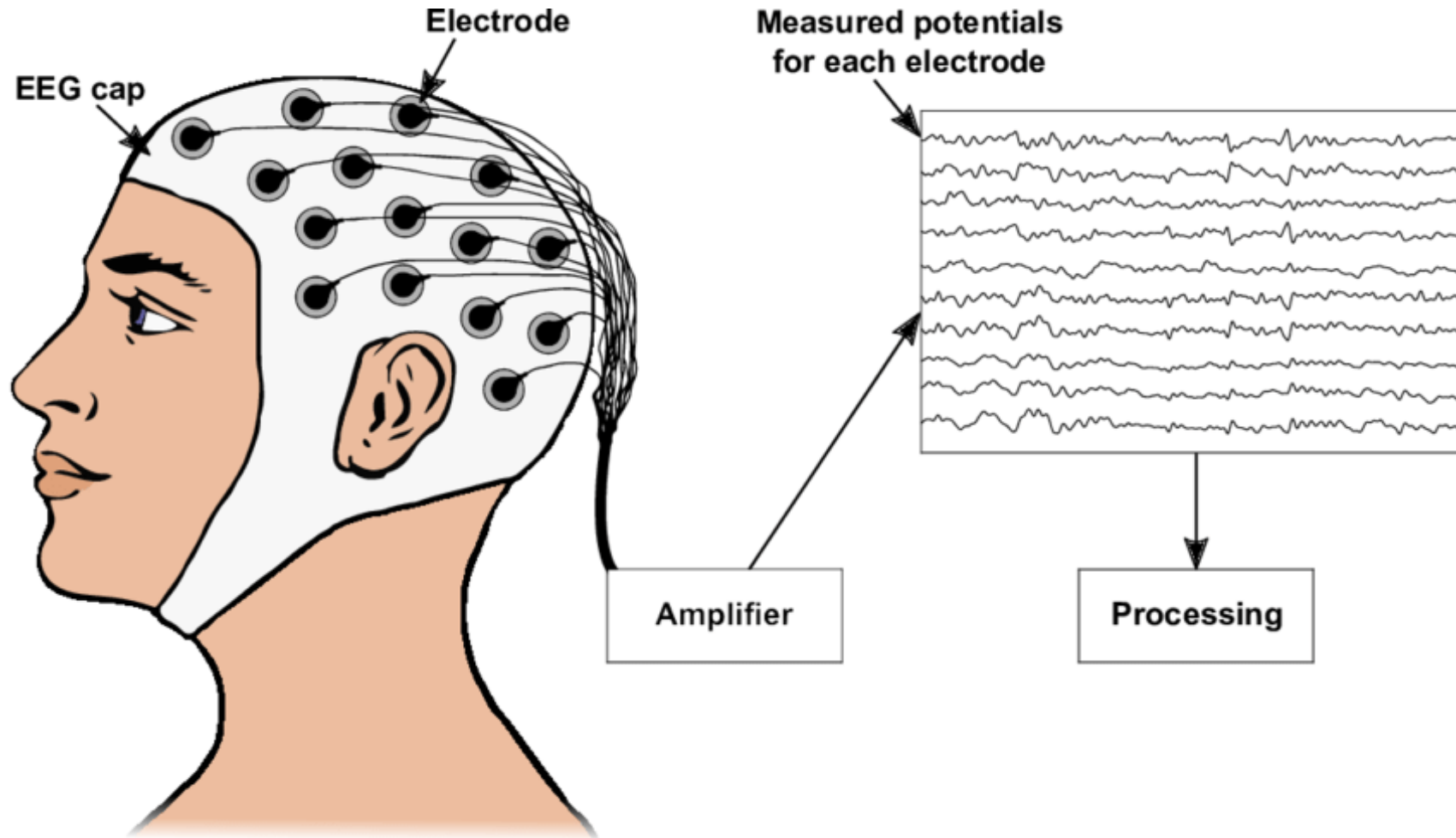




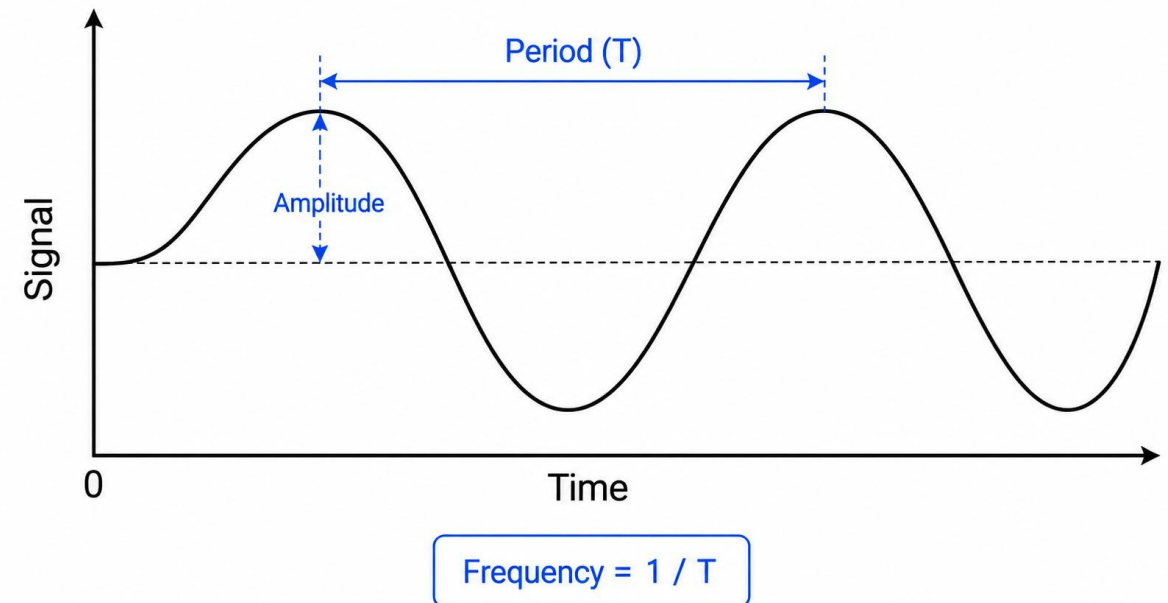
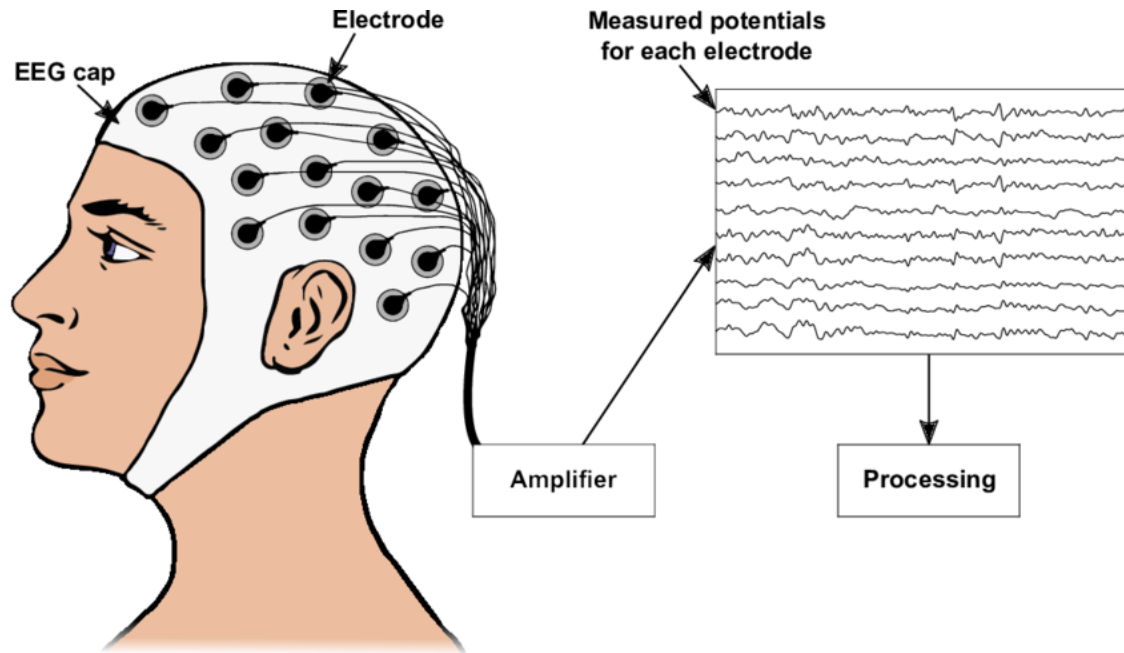
# RECORDING ELECTRICAL ACTIVITY OF THE BRAIN

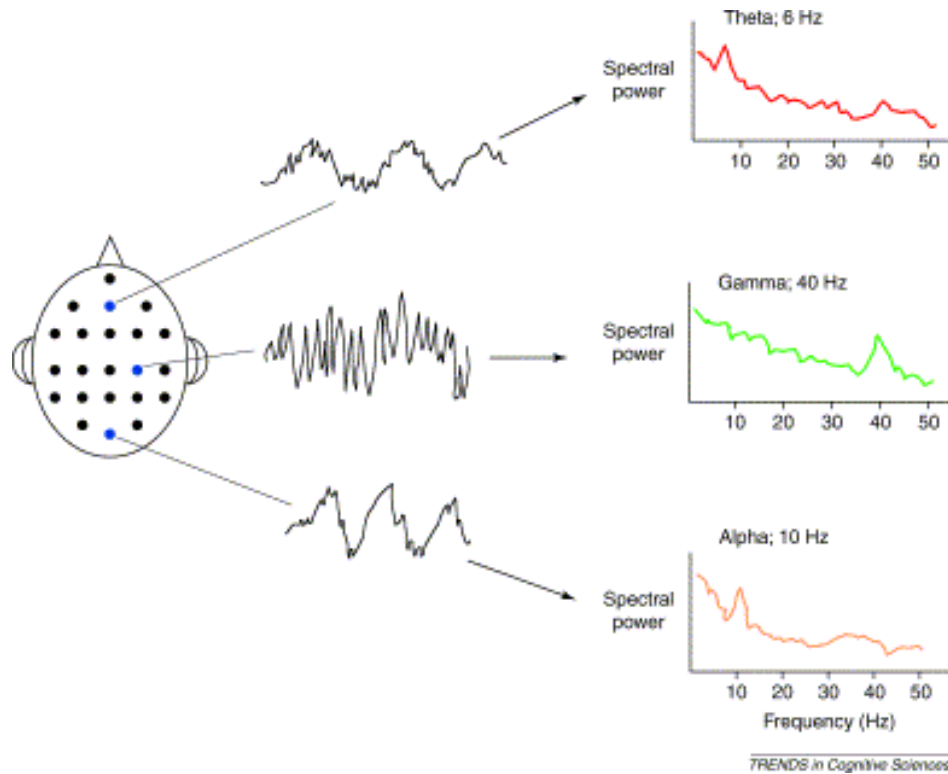


# ELECTROENCEPHALOGRAPHY (EEG)



# ELECTROENCEPHALOGRAPHY (EEG)

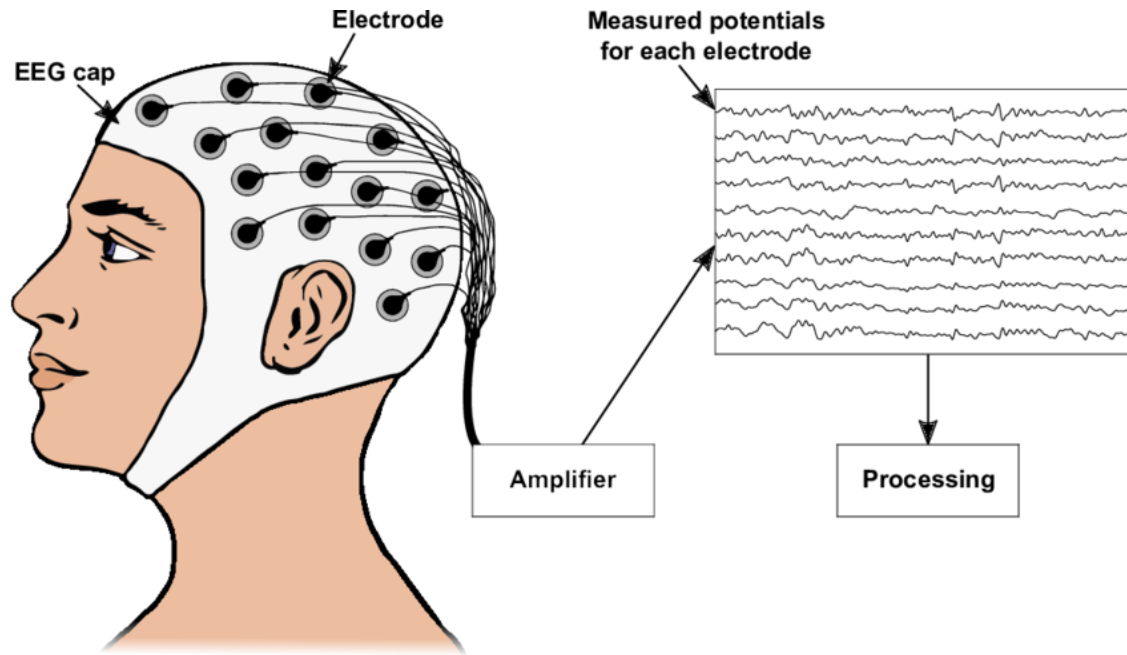




| Banda de frecuencias | Frecuencia | Estados cerebrales             |
|----------------------|------------|--------------------------------|
| Gamma ( $\gamma$ )   | >35 Hz     | Concentration, problem solving |
| Beta ( $\beta$ )     | 12-35 Hz   | Active mind, busy              |
| Alpha ( $\alpha$ )   | 8-12 Hz    | Restful, reflective            |
| Theta ( $\theta$ )   | 4-8 Hz     | Inward focused, drowsiness     |
| Delta ( $\delta$ )   | 0.5-4 Hz   | Sleep, dreaming                |

# BRAIN OSCILLATIONS

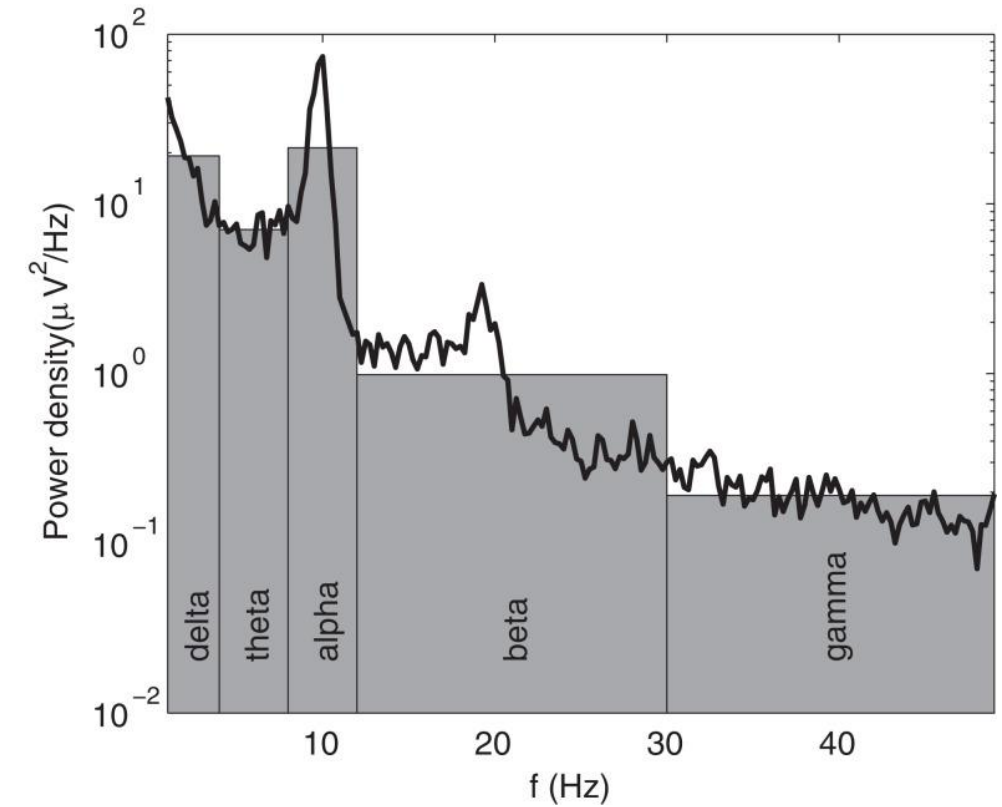
TIME DOMAIN



Fourier transform

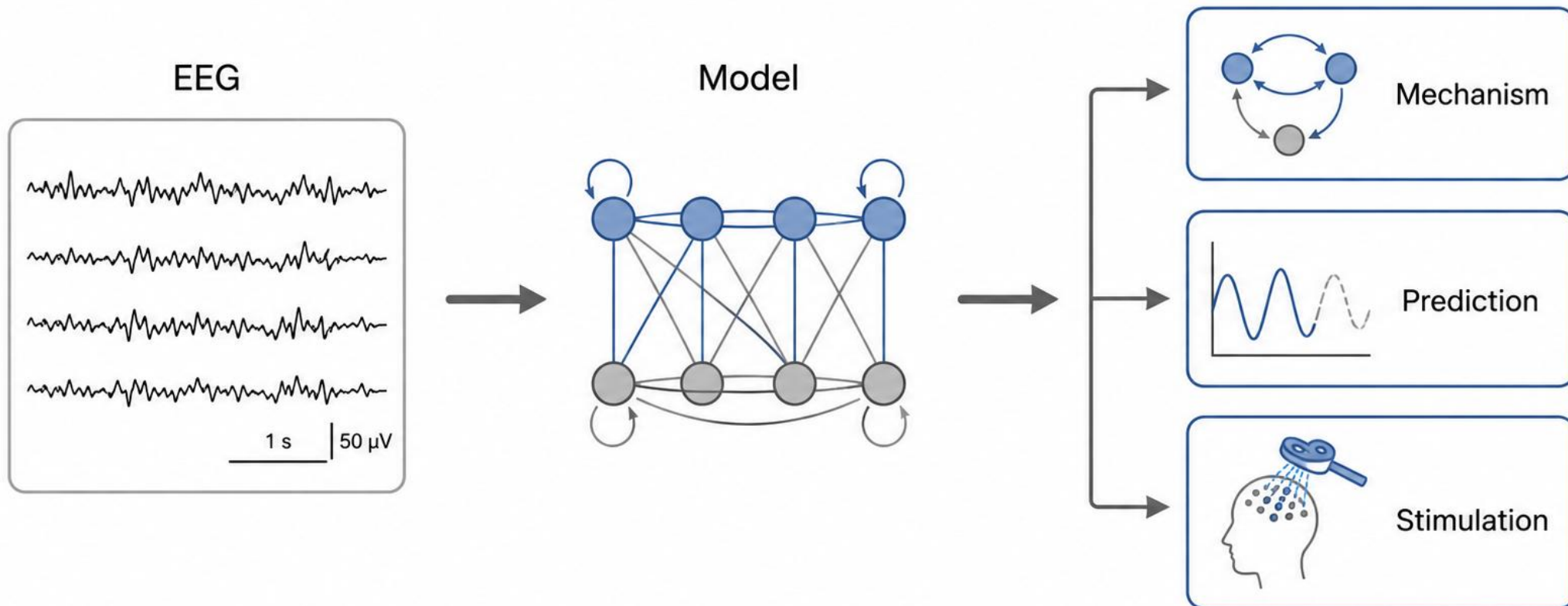


FREQUENCY DOMAIN



# WHY MODEL BRAIN RHYTHMS?

EEG tells us what happens.  
Models help us ask how and why.



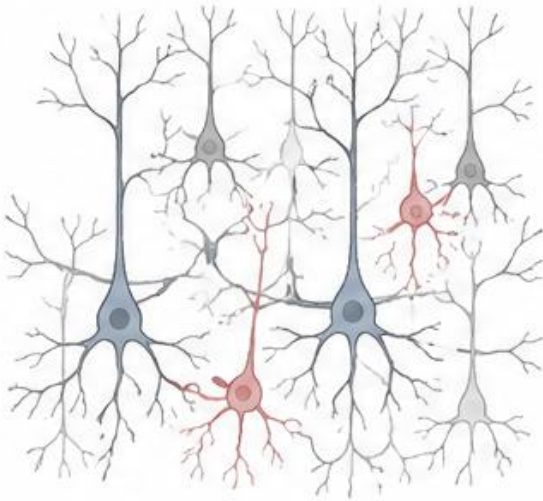


# NEURAL MASS MODELS: from many neurons to few variables

Neural mass models describe the **average (mean) activity** of interacting populations of neurons.  
They provide a bridge between **microscopic** neurons and **macroscopic** signals like EEG.

## 1. Microscopic level

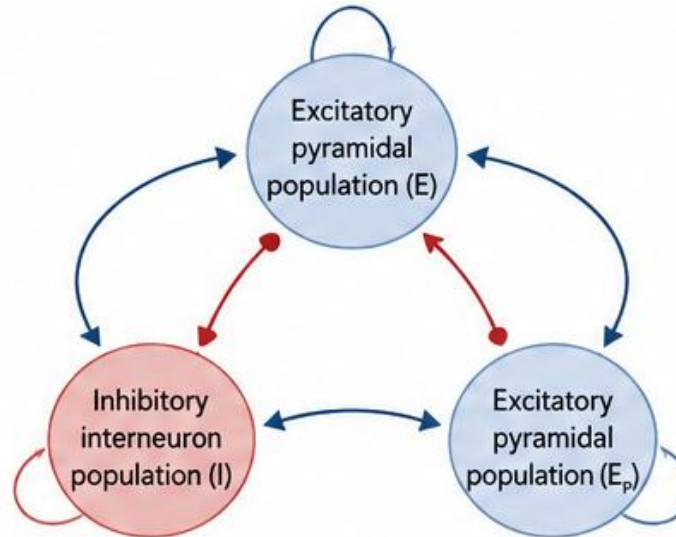
*Individual neurons*



Many neurons and synapses  
with complex dynamics

## 2. Mesoscopic level (Neural mass)

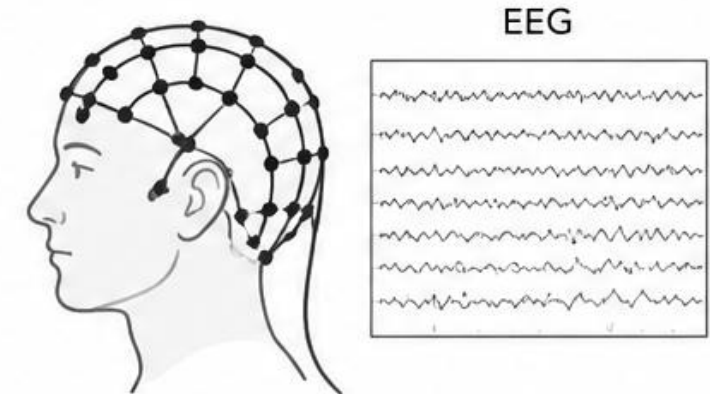
*Populations*



Average activity of each population  
described by a few variables

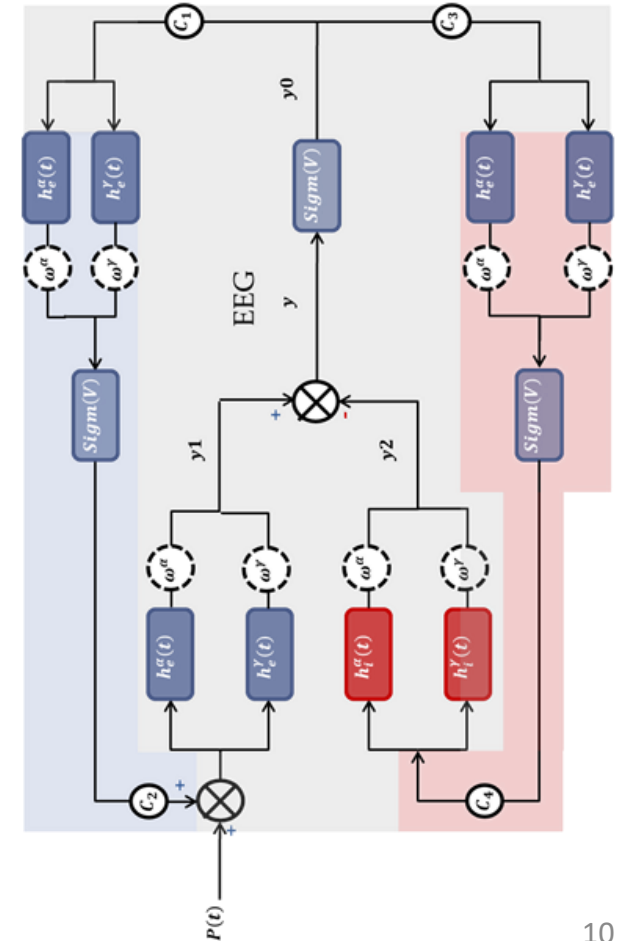
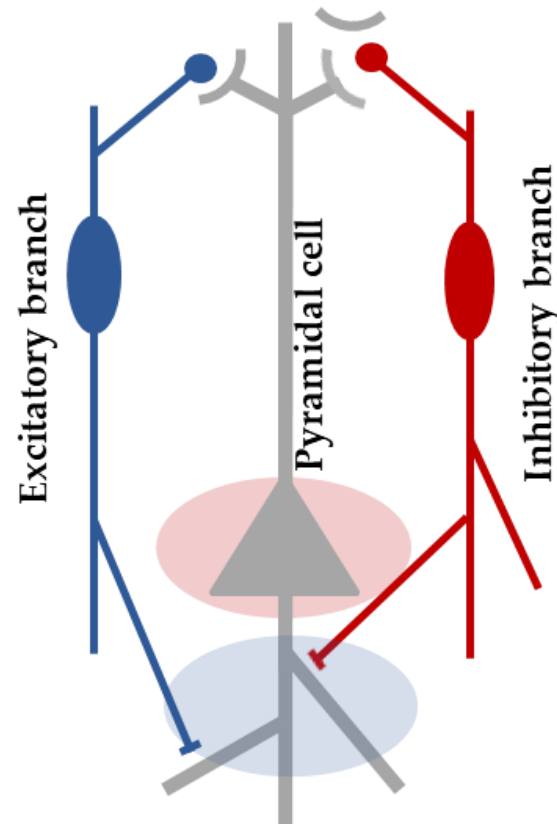
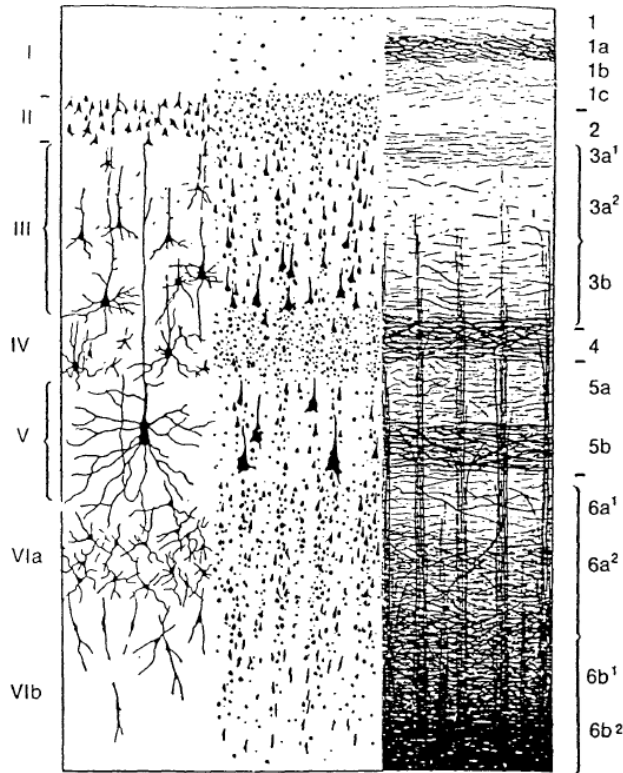
## 3. Macroscopic level

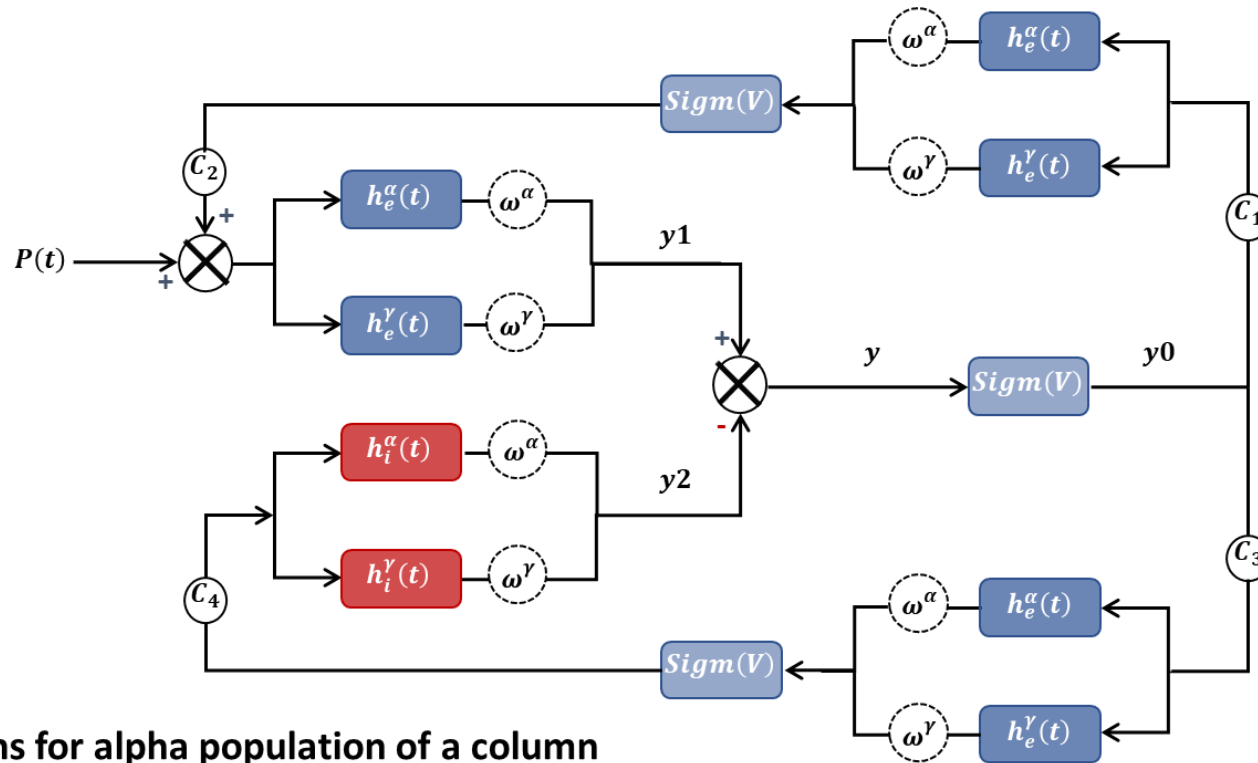
*Large-scale signals*



Summed electrical activity of very large  
populations recorded at the scalp

## Jansen-Rit Model: What Happens Inside One Cortical Column?





## Equations for alpha population of a column

$$\begin{aligned}\ddot{y}0^\alpha(t) &= A^\alpha a^\alpha \text{Sigm}(y1(t) - y2(t)) - 2a^\alpha \dot{y}0^\alpha(t) - (a^\alpha)^2 y0^\alpha(t) \\ \ddot{y}1^\alpha(t) &= A^\alpha a^\alpha \{P(t) + C_2 \text{Sigm}(C_1 y0(t))\} - 2a^\alpha \dot{y}1^\alpha(t) - (a^\alpha)^2 y1^\alpha(t) \\ \ddot{y}2^\alpha(t) &= B^\alpha b^\alpha C_4 \text{Sigm}(C_3 y0(t)) - 2b^\alpha \dot{y}2^\alpha(t) - (b^\alpha)^2 y2^\alpha(t)\end{aligned}$$

$h(t)$  — Synaptic filtering

*Firing rate  $\rightarrow$  postsynaptic potential*

$\text{Sigm}(v)$  — Population response

*Membrane potential  $\rightarrow$  firing rate*

## Constraint:

$$\omega^\alpha + \omega^\gamma = 1$$

$$\omega = \omega^\alpha \text{ then } \omega^\gamma = 1 - \omega$$

$$y0(t) = \omega y0^\alpha(t) + (1 - \omega) y0^\gamma(t)$$

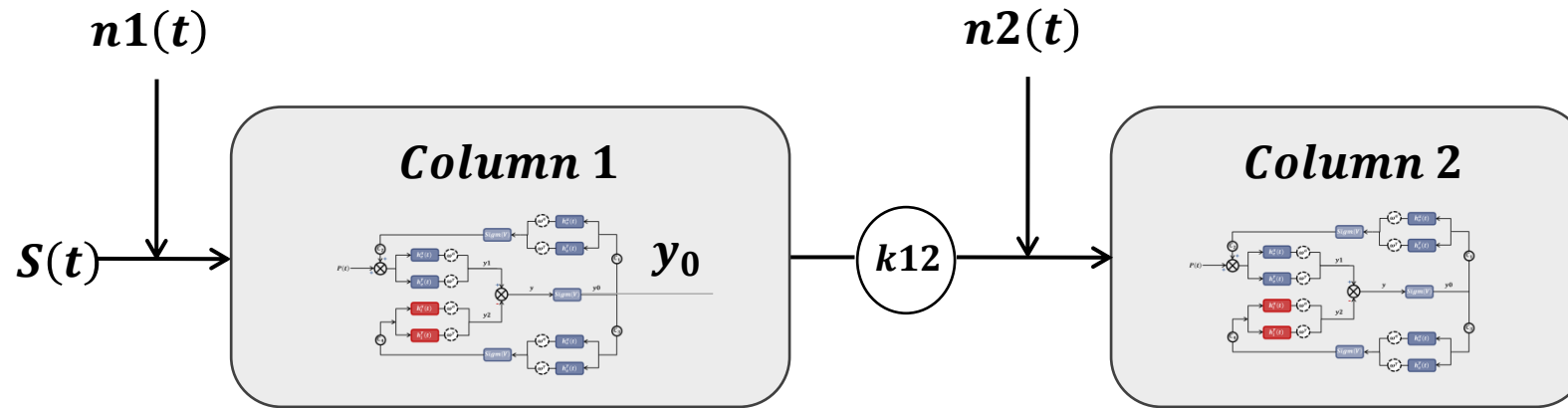
$$y1(t) = \omega y1^\alpha(t) + (1 - \omega) y1^\gamma(t)$$

$$y2(t) = \omega y2^\alpha(t) + (1 - \omega) y2^\gamma(t)$$

$$EEG_{like} y(t) = \omega(y1^\alpha - y2^\alpha) + (1 - \omega)(y1^\gamma - y2^\gamma)$$

## Coupling two Jansen–Rit neural masses

Double column double population model



Activity generated in one cortical region can influence the dynamics of another.

**local dynamics + connectivity = network dynamics.**



> [Biomed Phys Eng Express](#). 2022 Jun 28;8(4). doi: 10.1088/2057-1976/ac605a.

## Modelling neural entrainment and its persistence: influence of frequency of stimulation and phase at the stimulus offset

Mónica Otero <sup>1 2</sup>, Caroline Lea-Carnall <sup>3</sup>, Pavel Prado <sup>4</sup>, María-José Escobar <sup>5</sup>,  
Wael El-Deredy <sup>1 2 3</sup>

Affiliations + expand

PMID: 35320793 DOI: [10.1088/2057-1976/ac605a](https://doi.org/10.1088/2057-1976/ac605a) [↗](#)

### Abstract

Neural entrainment, the synchronization of brain oscillations to the frequency of an external stimuli, is a key mechanism that shapes perceptual and cognitive processes. *Objective.* Using simulations, we investigated the dynamics of neural entrainment, particularly the period following the end of the stimulation, since the persistence (reverberation) of neural entrainment may condition future sensory representations based on predictions about stimulus rhythmicity. *Methods.* Neural entrainment was

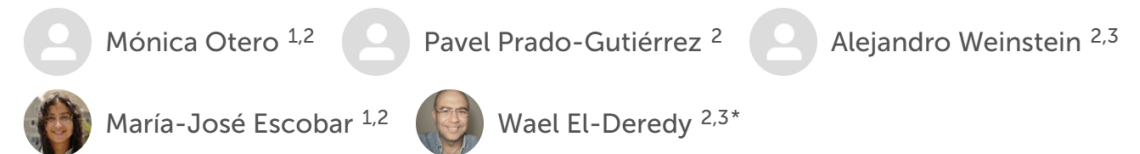
### ORIGINAL RESEARCH article

Front. Hum. Neurosci., 08 April 2020

Sec. Sensory Perceptual Neuroscience

Volume 14 - 2020 | <https://doi.org/10.3389/fnhum.2020.00139>

## Persistence of EEG Alpha Entrainment Depends on Stimulus Phase at Offset

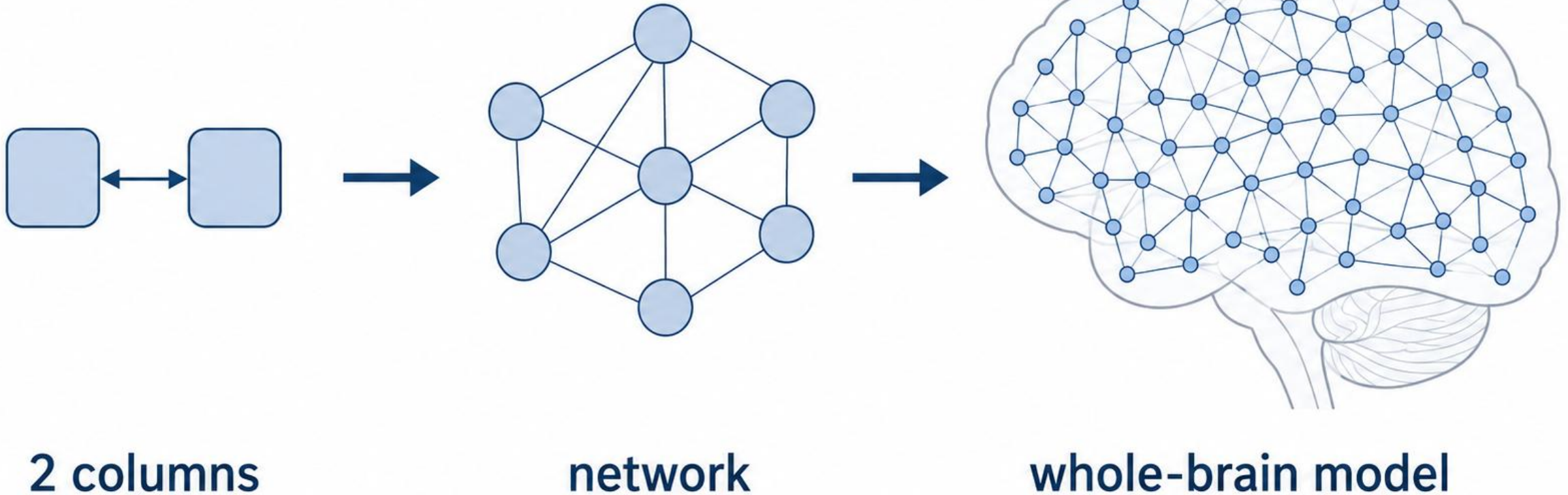


1. Department of Electronic Engineering, Universidad Técnica Federico Santa María, Valparaíso, Chile

2. Advanced Center for Electrical and Electronic Engineering (AC3E), Universidad Técnica Federico Santa María, Valparaíso, Chile [See more](#)

# FROM TWO COLUMNS TO THE WHOLE BRAIN

Whole-brain modeling = local dynamics + brain connectivity

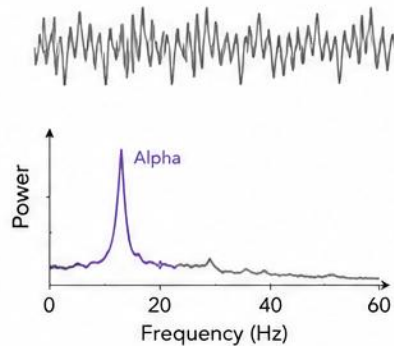


# WHY MODEL BRAIN OSCILLATIONS?

*From description to mechanism, from local to whole-brain*

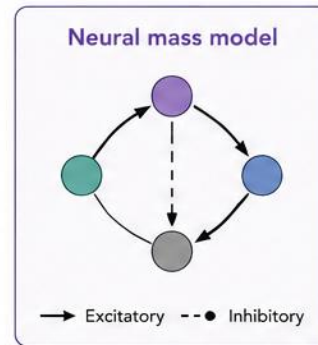
## 1. WE OBSERVE

EEG reveals rhythms and their frequencies

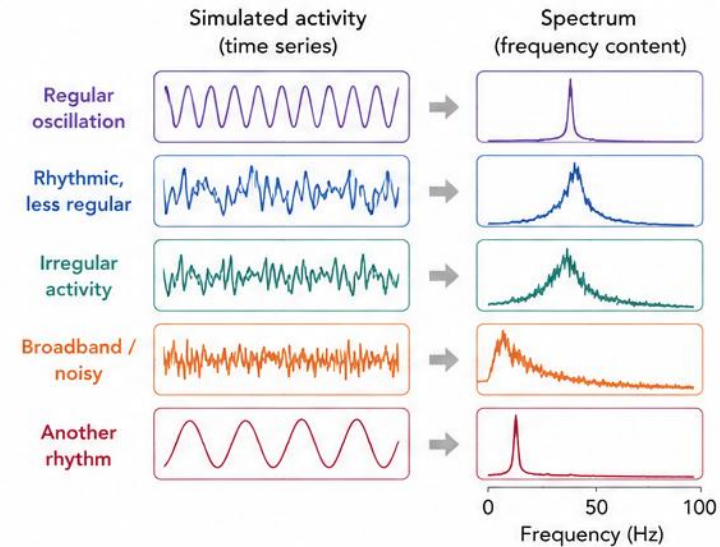


## 2. WE MODEL

Models test mechanisms and generate rhythms

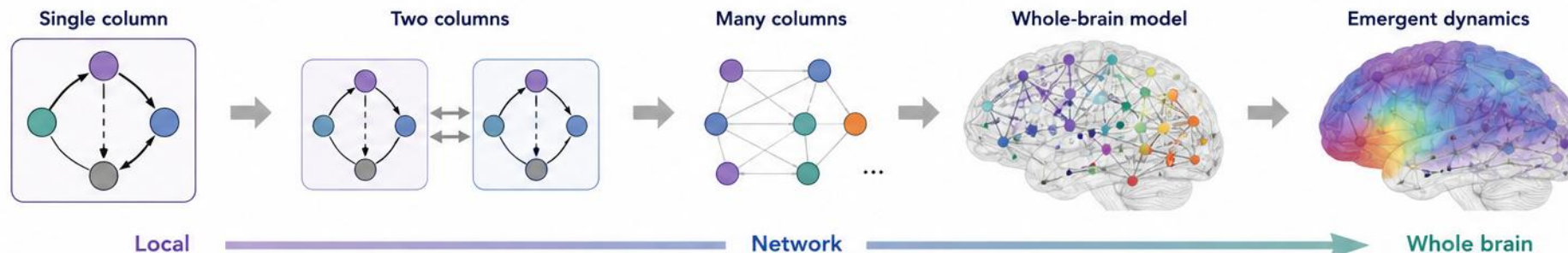


## 3. DIFFERENT RHYTHMIC REGIMES EMERGE



Changing model parameters or inputs → different rhythms

## FROM A LOCAL MODEL TO WHOLE-BRAIN NETWORKS



MODELS ALLOW US TO MOVE FROM OBSERVING BRAIN RHYTHMS TO TESTING MECHANISMS THAT COULD GENERATE THEM.

## COUPLED OSCILLATORS



## KURAMOTO MODEL

$$\dot{\theta}_n(t) = \omega_n + \frac{K}{N} \sum_{m=1}^N C_{nm} \sin(\theta_m(t) - \theta_n(t))$$

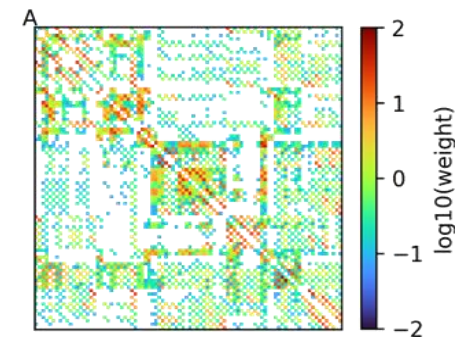
Global coupling strength

Weight of connection

Natural frequency

Interaction term

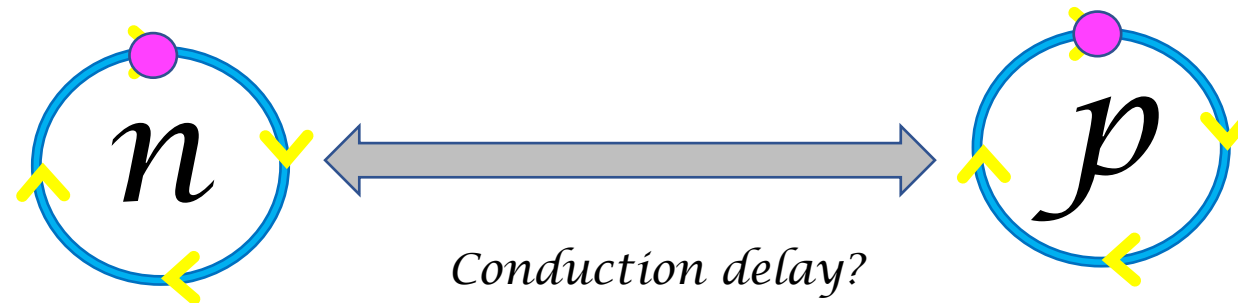
## WEIGHTS



Weights are estimated from the number of tracts connecting two brain regions



## COUPLED OSCILATORS WITH DELAYS

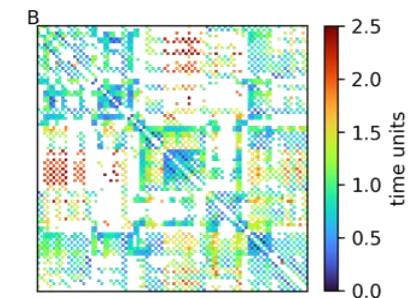


## KURAMOTO MODEL WITH DELAYS

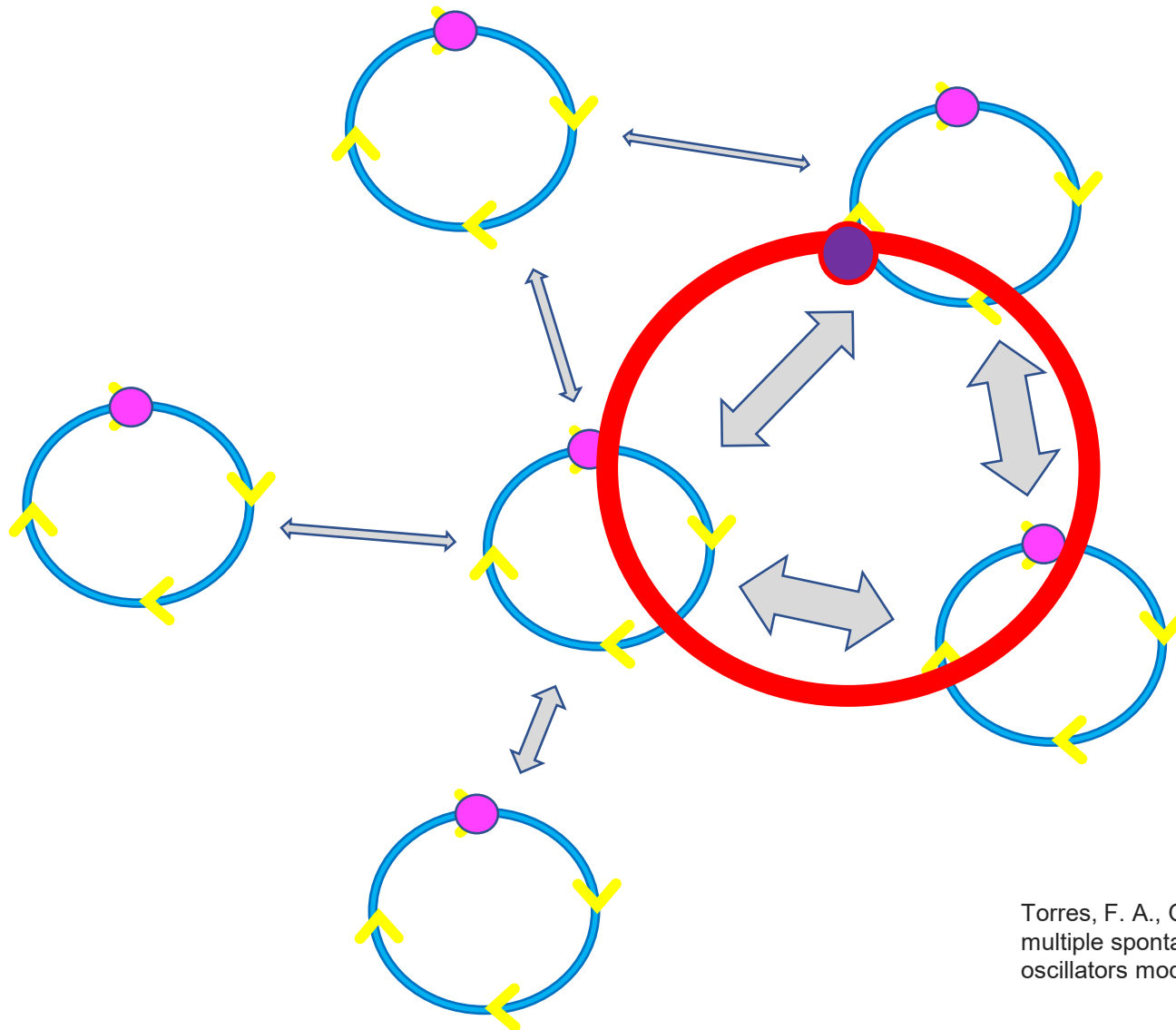
$$\dot{\theta}_n(t) = \omega_n + \frac{K}{N} \sum_{m=1}^N c_{nm} \sin(\theta_m(t - \tau_{nm}) - \theta_n(t))$$

Conduction delay

## DELAYS

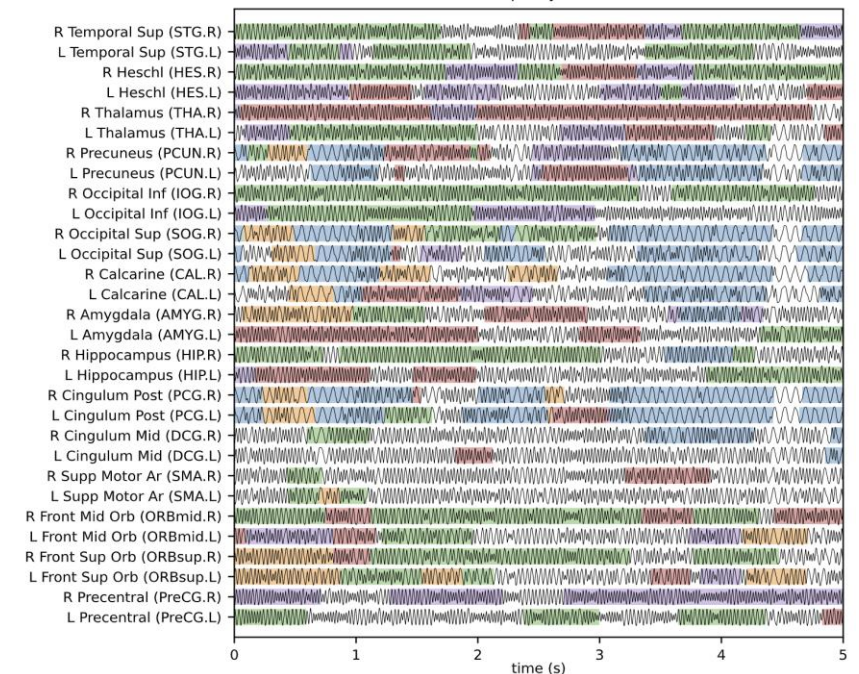


# MODELLING BRAIN OSCILLATIONS



Given a configuration of parameters (K and MD):

- If coupling is sufficiently strong subgroups of units will synchronize.
- Due to delays, nodes dynamics will switch between states (frequencies).



Torres, F. A., Otero, M., Lea-Carnall, C. A., Cabral, J., Weinstein, A., & El-Deredy, W. (2024). Emergence of multiple spontaneous coherent subnetworks from a single configuration of human connectome coupled oscillators model. *Scientific Reports*, 14(1), 30726.

Article | [Open access](#) | Published: 28 December 2024

## Emergence of multiple spontaneous coherent subnetworks from a single configuration of human connectome coupled oscillators model

[Felipe A. Torres](#), [Mónica Otero](#), [Caroline A. Lea-Carnall](#), [Joana Cabral](#), [Alejandro Weinstein](#) & [Wael El-Deredy](#) 

[Scientific Reports](#) **14**, Article number: 30726 (2024) | [Cite this article](#)

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JOURNAL ARTICLE

## Reductive Perturbation Approach to Chemical Instabilities FREE

[Yoshiki Kuramoto](#), [Toshio Tsuzuki](#) | [Author Notes](#)

*Progress of Theoretical Physics*, Volume 52, Issue 4, October 1974, Pages 1399–1401,  
<https://doi.org/10.1143/PTP.52.1399>

**Published:** 01 October 1974 | [Article history](#) ▼

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## Role of local network oscillations in resting-state functional connectivity

[Joana Cabral](#) <sup>a 1</sup> , [Etienne Hugues](#) <sup>a 1</sup>, [Olaf Sporns](#) <sup>b</sup>, [Gustavo Deco](#) <sup>a c</sup> | [Show more](#) ▼

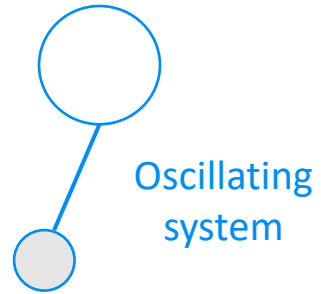
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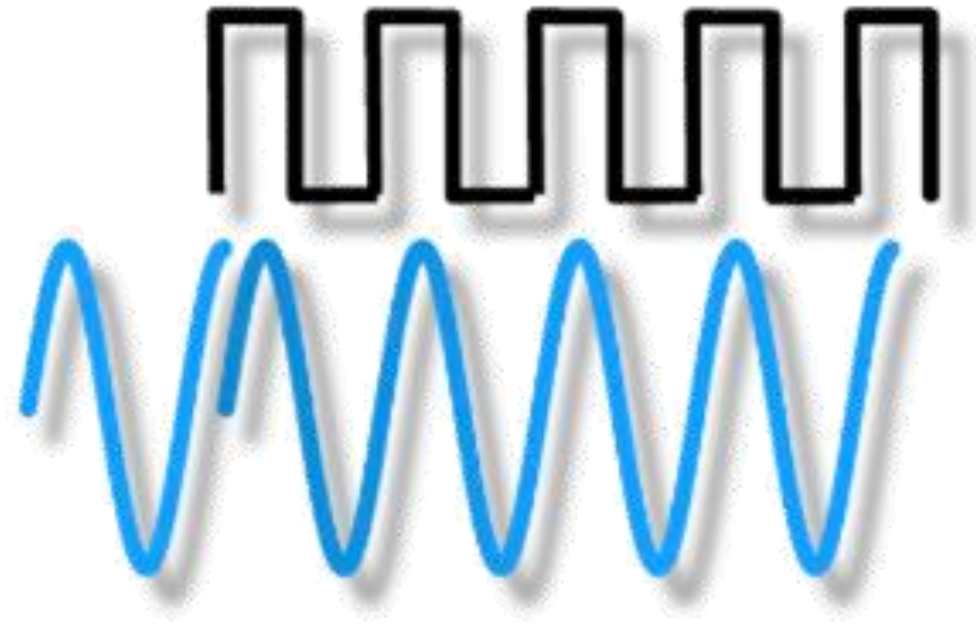
### Abstract

Spatio-temporally organized low-frequency fluctuations (<0.1 Hz), observed in BOLD fMRI signal during rest, suggest the existence of underlying network dynamics that emerge spontaneously from intrinsic brain processes. Furthermore, significant correlations between distinct anatomical regions—or functional connectivity (FC)—have led to the identification of several widely distributed resting-state networks (RSNs). This slow dynamics seems to be highly structured by anatomical connectivity but the mechanism behind it and its relationship with neural activity, particularly in the gamma frequency range, remains largely unknown. Indeed, direct measurements of neuronal activity have revealed similar large-scale correlations, particularly in slow power fluctuations of local field potential gamma frequency range

EXTERNAL RYTHMIC  
DRIVE



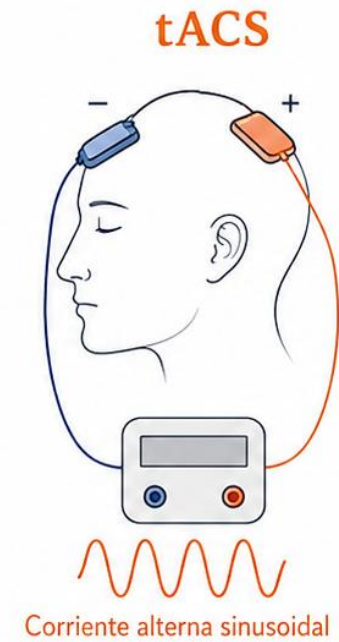
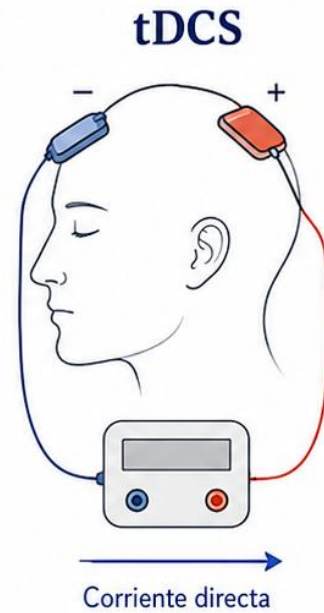
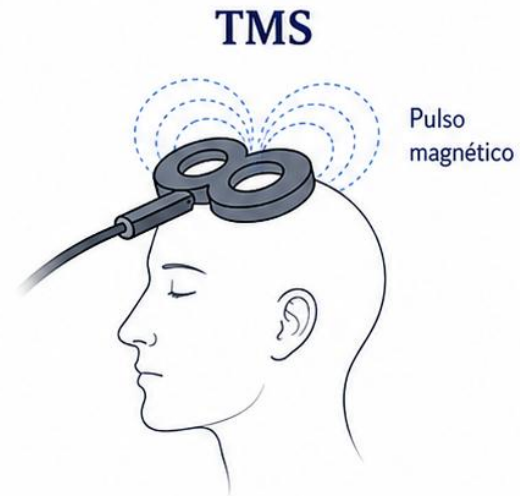
## Entrainment





# NON- INVASIVE NEUROSTIMULATION

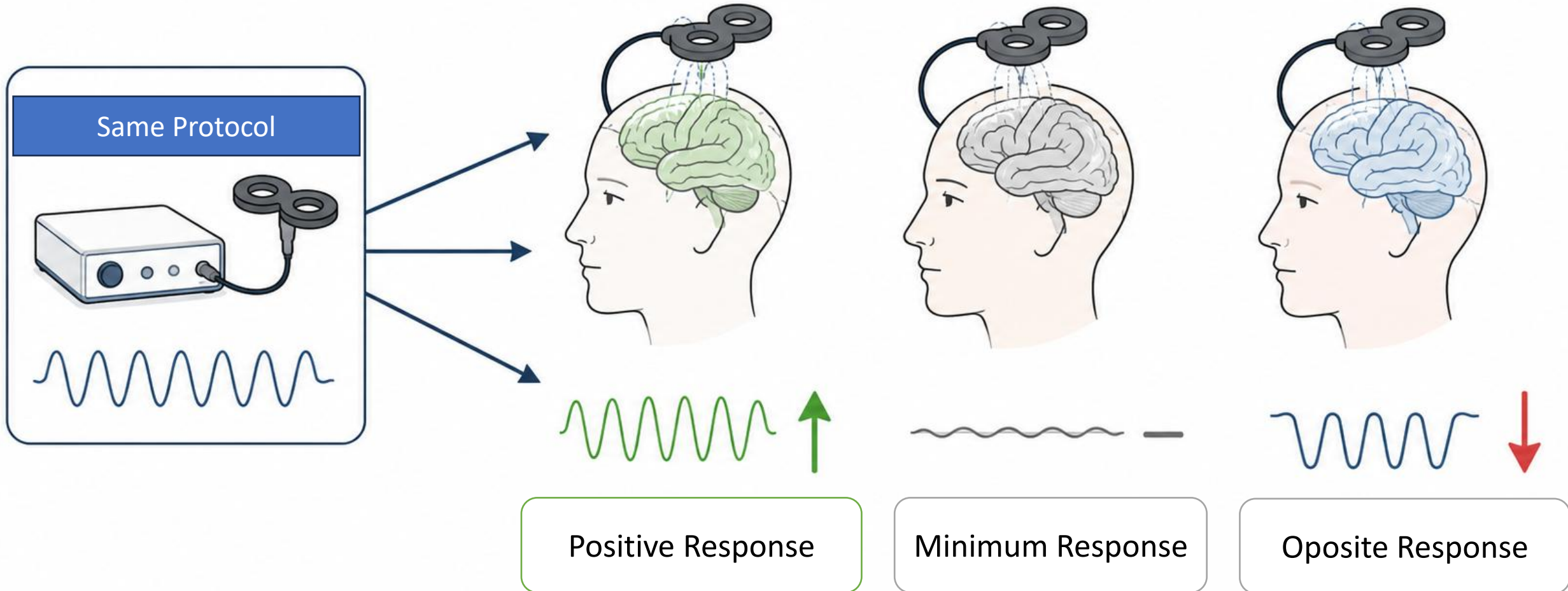
## Techniques



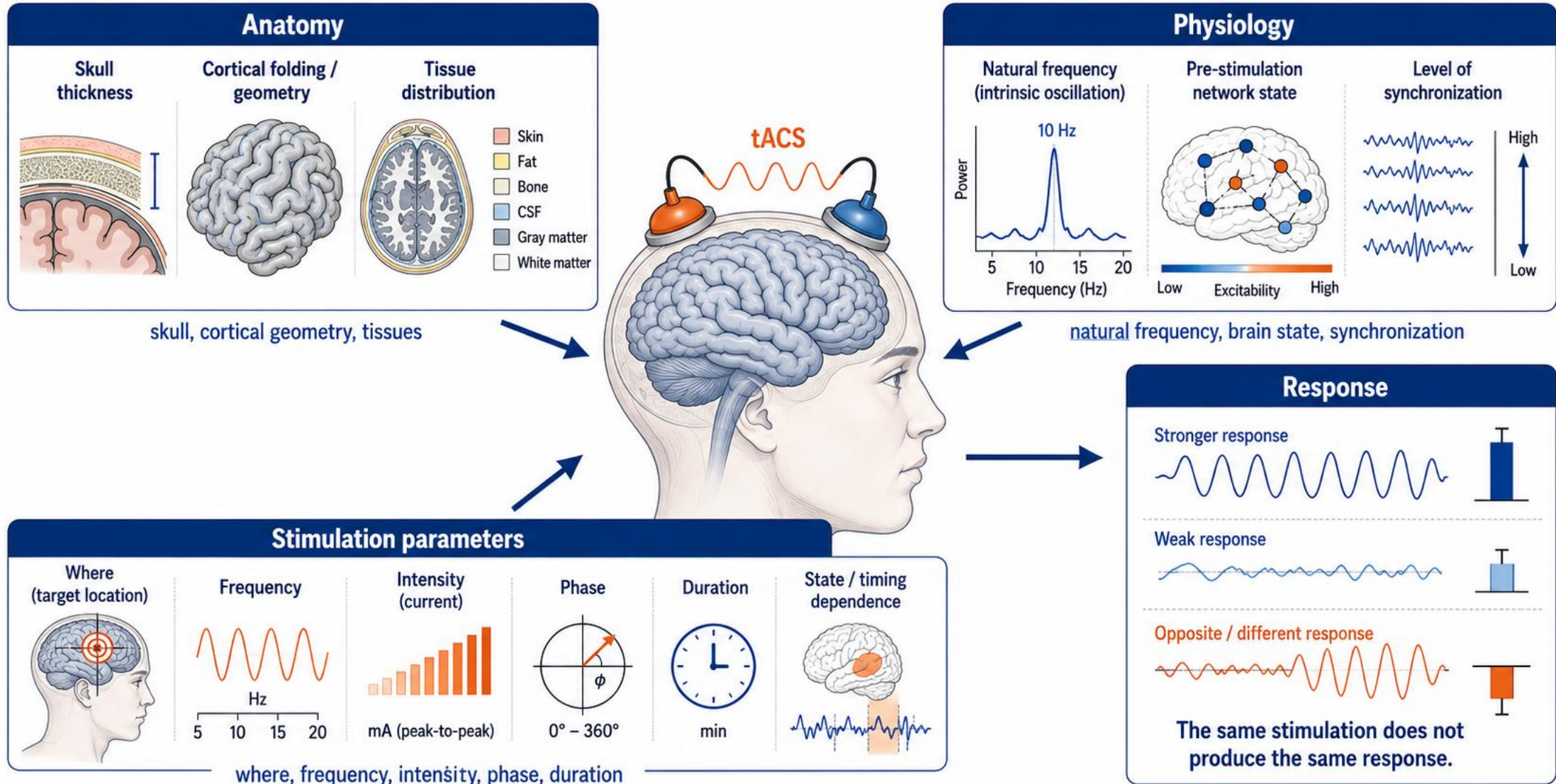
## Areas of research



# NEUROSTIMULATION: LARGE INTERINDIVIDUAL VARIABILITY



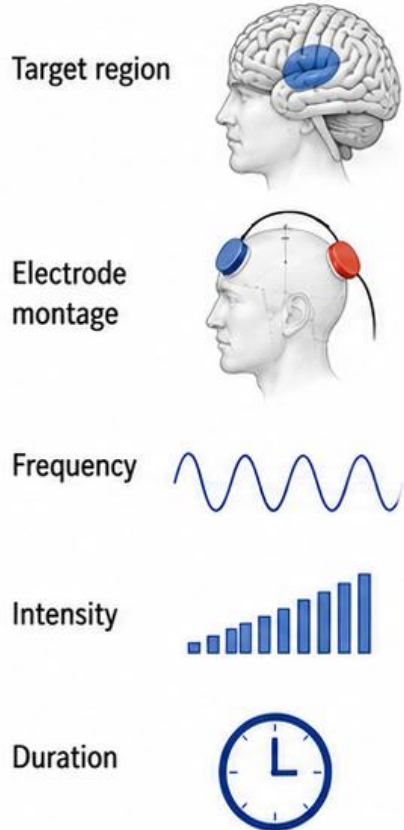
# WHERE DOES THE VARIABILITY COME FROM?



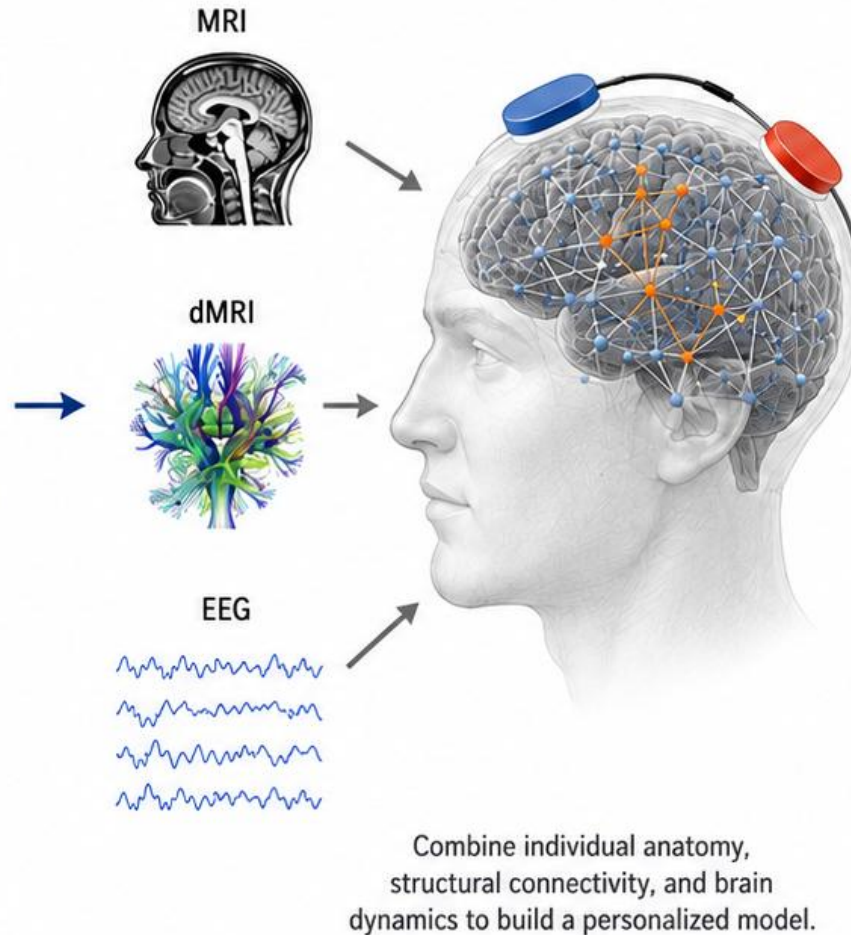


# FROM TRIAL-AND-ERROR TO MODEL-GUIDED STIMULATION

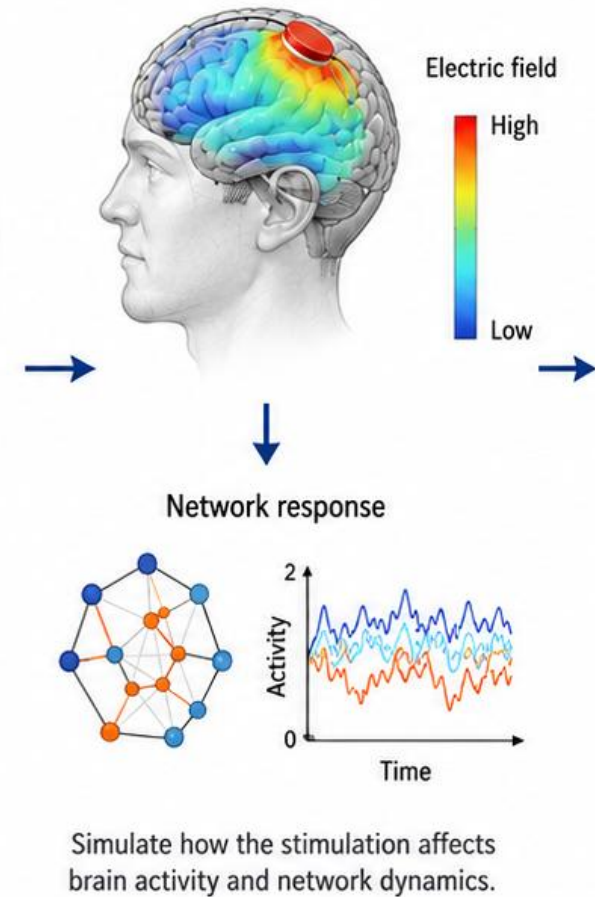
## Possible parameters



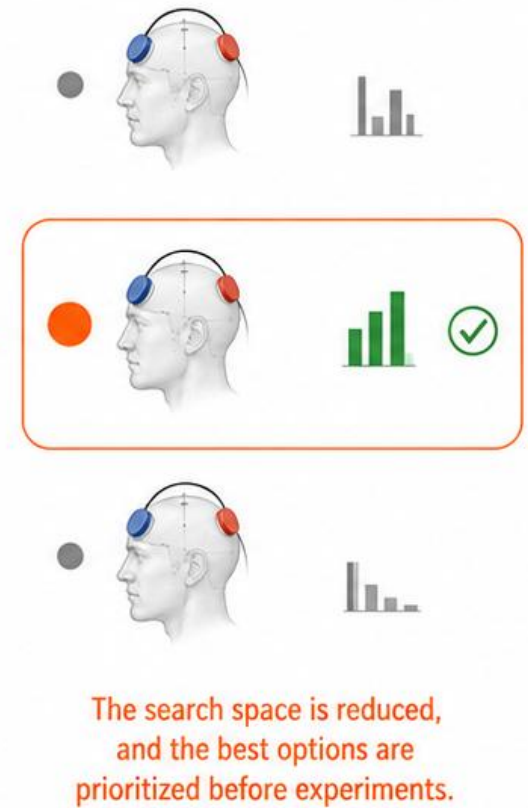
## Individual model



## Simulation

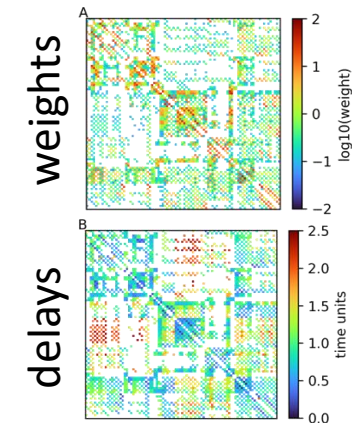
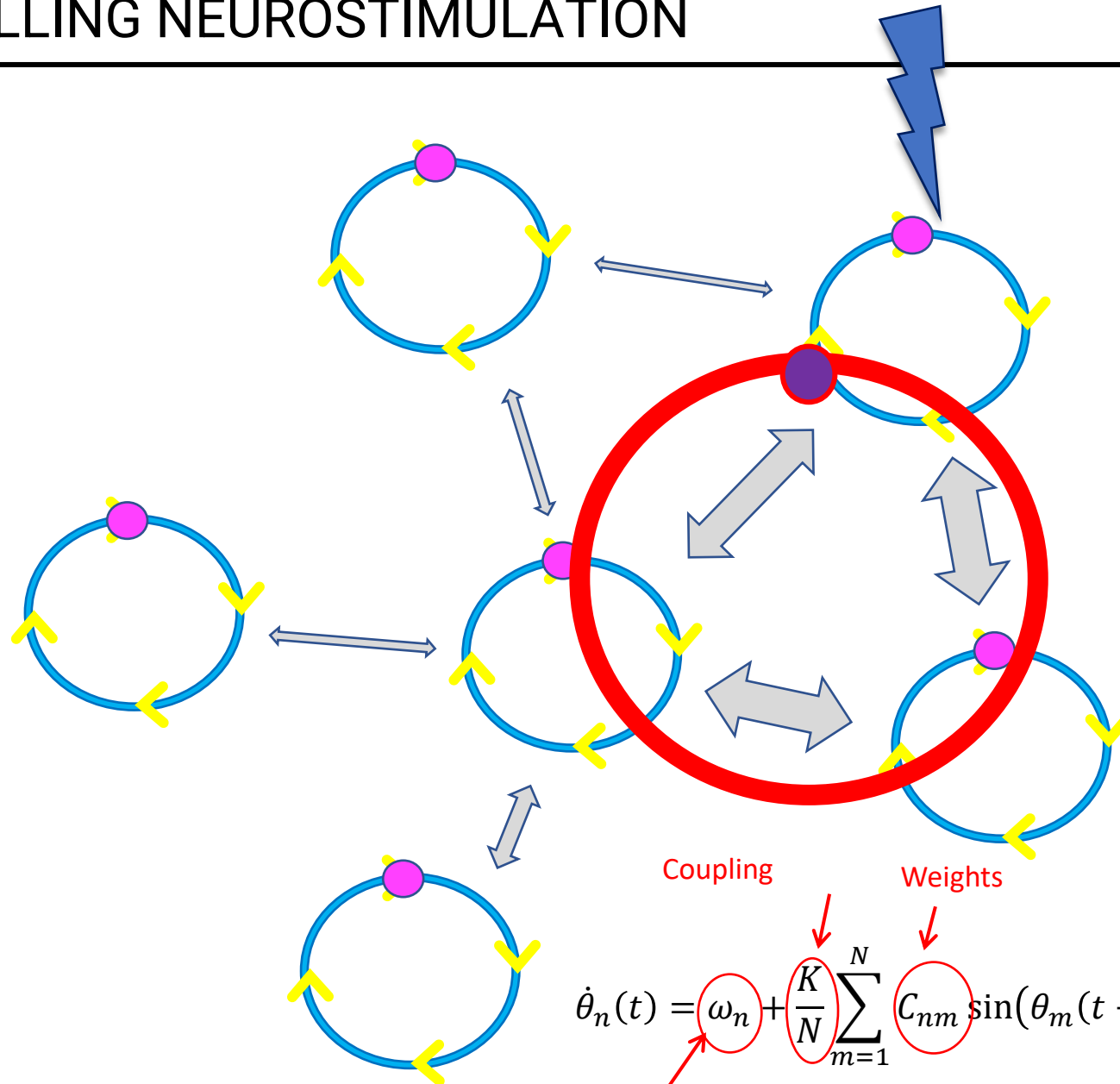


## Informed selection





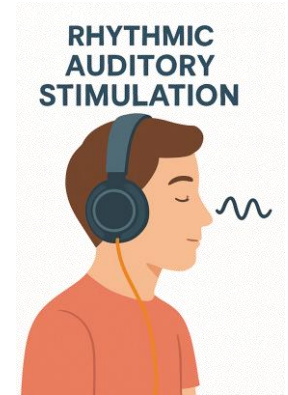
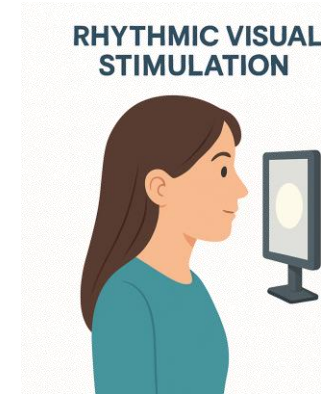
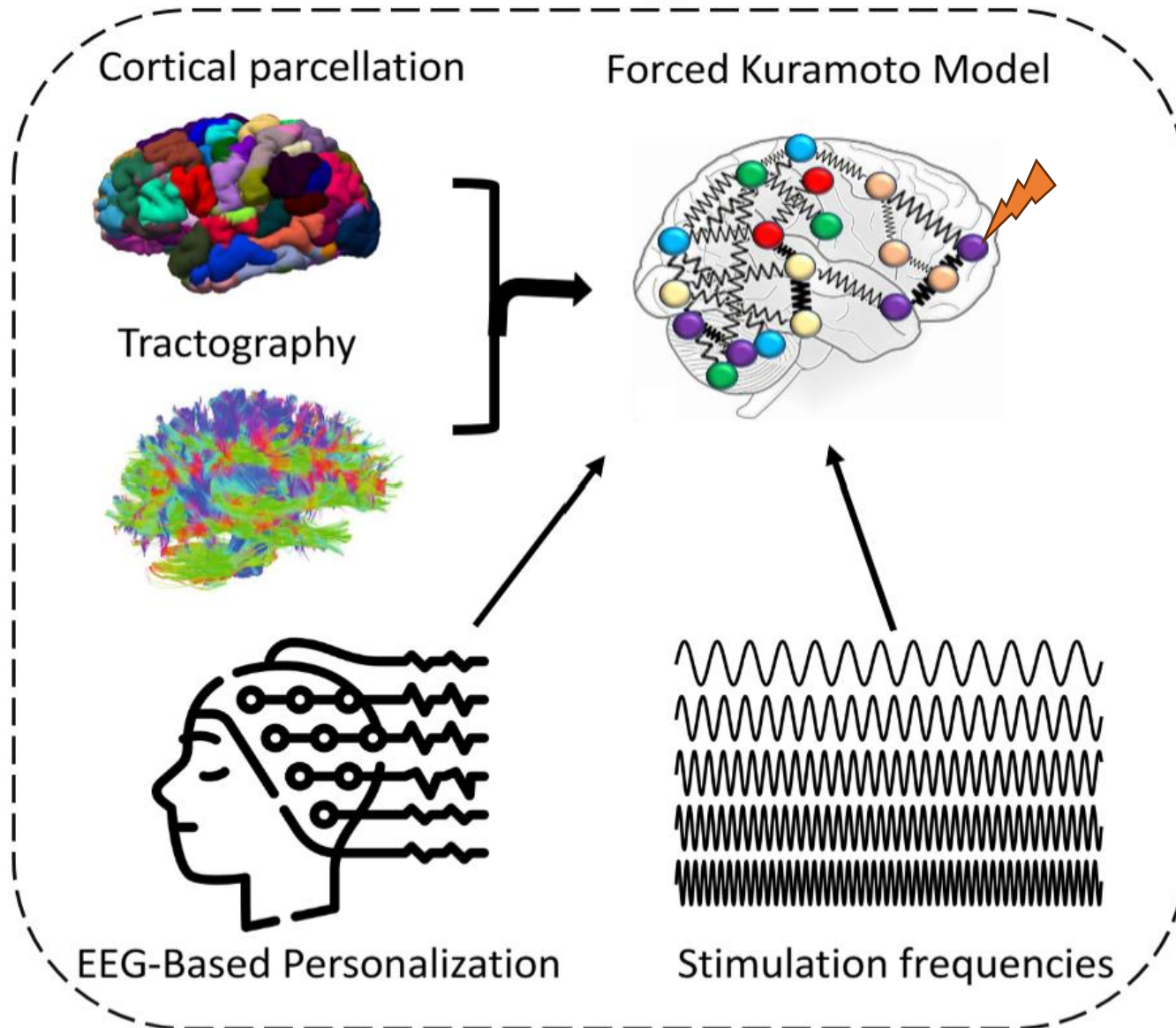
WHAT IS THE EFFECT OF AN EXTERNAL PERIODIC STIMULATION ?



*estimated from the number  
of tracts connecting two  
brain regions*

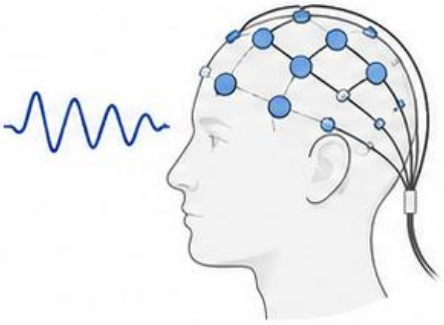
*computed from the  
average length of the  
fibers*

# MODELING SENSORY RHYTHMIC STIMULATION



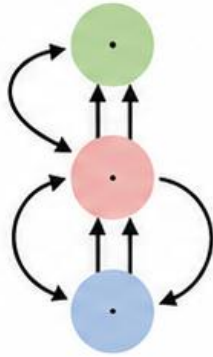
# TAKE-HOME MESSAGES

## 1 Brain rhythms



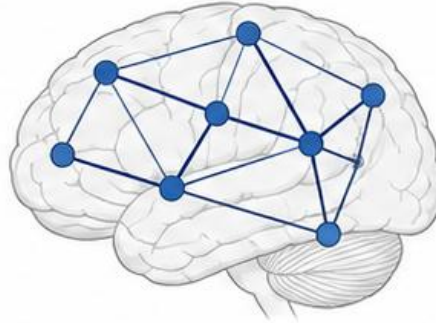
can be measured  
with EEG

## 2 Jansen-Rit



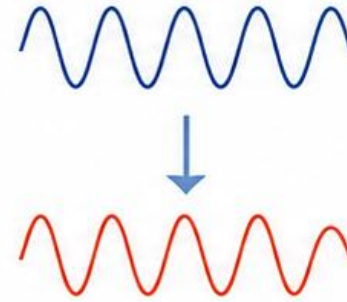
models local  
population dynamics

## 3 Oscillatory networks



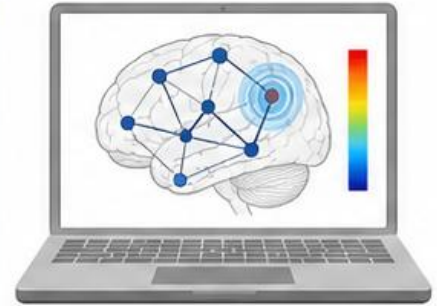
represent whole-brain  
interactions

## 4 Entrainment



external rhythms perturb  
ongoing dynamics

## 5 Model-guided stimulation



helps explore and  
personalize protocols

**Whole-brain models integrate local dynamics,  
connectivity, and stimulation.**



## Agradecimientos:

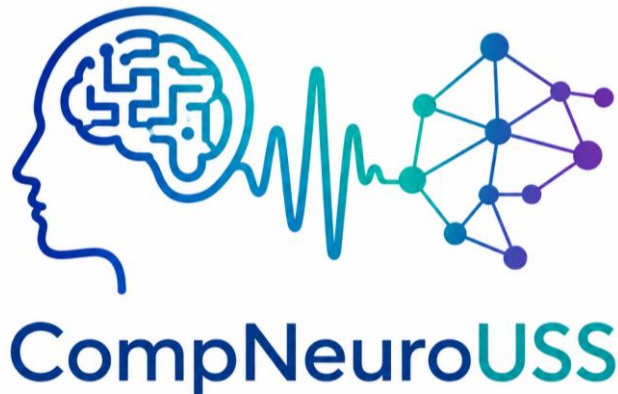
This work is supported by ANID, Chile:

- Fondecyt INICIACION 11241484
- Fondecyt EXPLORACION 13240042
- Centro Basal Ciencia & Vida, FB210008





UNIVERSIDAD  
SAN SEBASTIAN



# Lab in Computational Neuroscience

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**GRACIAS !!**